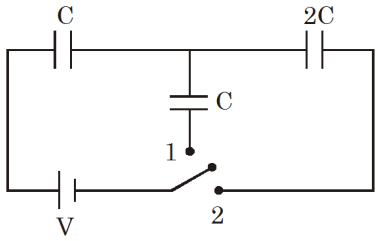
PART-1 : PHYSICS

SECTION-I

1) In the given circuit diagram, switch was connected to position 1 for long time. At t = 0, switch is shifted from position 1 to position 2. Find the final charge on capacitor 2C.



(A) $\frac{CV}{6}$

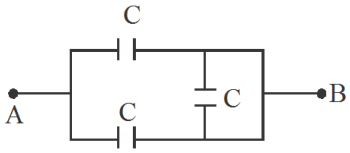
(B) $\frac{CV}{3}$

(C) $\frac{2CV}{3}$

(D) $\frac{4CV}{3}$

2)

Find equivalent capacity of the combination between A and B.



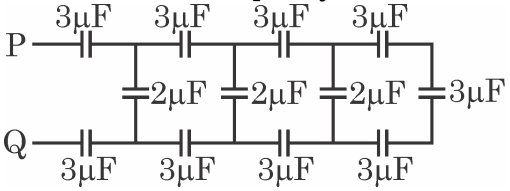
(A) $\frac{C}{2}$

(B) $\frac{3C}{2}$

(C) 2C

(D) $\frac{2C}{3}$

3) The effective capacity in the following figure between the points P and Q will be :



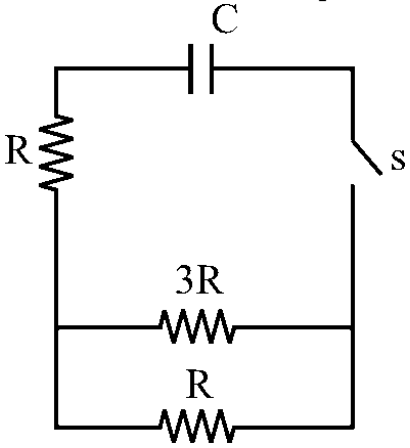
(A) 3μF

(B) 5μF

(C) 2μF

(D) 1μF

4) In the figure, capacitor is completely charged and switch is closed at t = 0. The time after which the current from the capacitor becomes $1/4^{th}$ of its maximum value will be :-



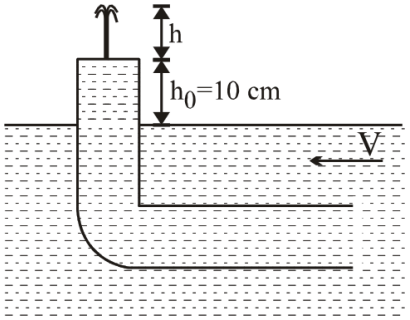
(A) 2 RC ln2

(B) $\frac{7RC}{4}$ ln2

(C) RC ln2

(D) $\frac{7RC}{2}$ ln2

5) A bent tube is lowered into water stream as shown in the figure. The velocity of the stream relative to the tube is equal to V = 2 m/s. The closed upper end of the tube located at height h_0 = 10 cm above free surface of water has a small orifice. To what height h will the water get spurt ?



(A) 5 cm

(B) 10 cm

(C) 20 cm

(D) 40 cm

6) There is a horizontal film of soap solution. On it a thread is placed in the form of a loop. The film is punctured inside the loop and the thread becomes a circular loop of radius R. If the surface tension of the soap solution be T, then the tension in the thread will be

(A) $\pi R^2/T$

(B) $\pi^2R^2T^2$

(C) $2\pi RT$

(D) $2RT$

7) Two drops of same radius are falling through air with steady velocity of v cm/s. If the two drops coalesce, what would be the terminal velocity?

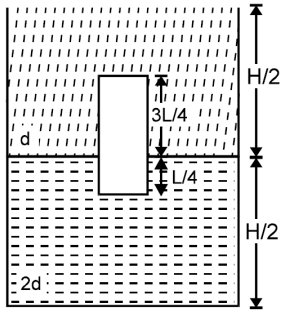
(A) 4 v

(B) $(4)^{1/3}$ v

(C) 2 v

(D) 64 v

8) A container of a large uniform cross-sectional area A resting on a horizontal surface holds two immiscible, non-viscous and incompressible liquids of densities ' d ' and ' 2 d ' each of height (1/2)H as shown. The smaller density liquid is open to atmosphere. A homogeneous solid cylinder of length $L\left(<\frac{1}{2}H\right)$ and cross-sectional area (1/5) A is immersed such that it floats with its axis vertical to the liquid-liquid interface with length (1/4) L in denser liquid. If D is the density of the solid cylinder then :

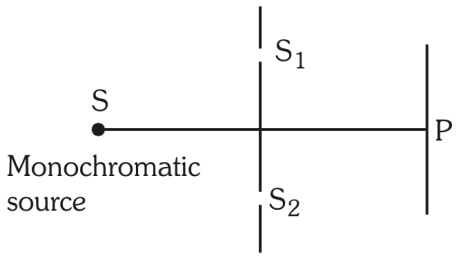


(A) $D = \frac{3d}{2}$

(B) $D = \frac{d}{2}$

(C) $D = \frac{2d}{3}$

(D) $D = \frac{5d}{4}$



9) A Young's double slit appratus is shown in figure.

	List-I		List-II
(P)	Source S move upwards	(1)	Central fringe moves upwards
(Q)	A thin glass slab is placed in front of slit S_2	(2)	Central fringe moves downward
(R)	Whole apparatus is immersed in water	(3)	Fringe width remains constant

(S)	Diameter of slit S_2 is doubled	(4)	Fringe width decreases
		(5)	Intensity of central maxima increases

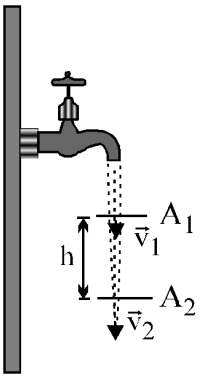
(A) P→1, Q→2, R→3, S→4

(B) P→2,3, Q→1,3, R→4, S→5

(C) P→2, Q→3, R→3, S→4

(D) P→1, Q→2, R→2, S→3

10) A steady stream of water falls straight down from a pipe. Assume the flow is incompressible ; the flow is similar to figure. How does the pressure in the water vary with height in the falling stream ?



(A) The pressure in the water is higher at lower points in the stream

(B) The pressure in the water is lower at lower points in the stream

(C) The pressure in the water is the same at all point in the stream

(D) The pressure of water is maximum at the middle of the stream

11) The equation of a wave on a string having linear mass density 0.07 kgm^{-1} is given as

$$y = 0.03\,(m)\cos\left[2\pi\left(\frac{t}{0.08\,(s)} - \frac{x}{0.8\,(m)}\right)\right]$$

the tension in string is :

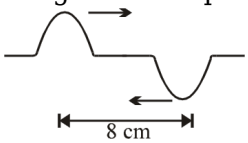
(A) 14N

(B) 3.5N

(C) 7 N

(D) 70N

12) Two pulses in a stretched string whose centres are initially 8 cm apart are moving towards each other as shown in figure. The speed of each pulse is 2 cm/s. After 2 seconds, the total energy of the pulses will be :



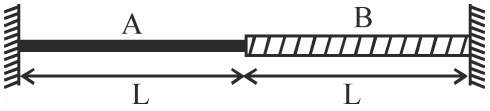
(A) zero

(B) purely kinetic

(C) purely potential

(D) partly kinetic and partly potential

13) A wire of length 2L, is made by joining two wires A and B of same length but different radii r and 2r and made of the same material. It is vibrating at a frequency such that the joint of the two wires forms a node. If the number of antinodes in wire A is p and that in B is q then the ratio p : q is :



(A) 4 : 9

(B) 3 : 5

(C) 1 : 4

(D) 1 : 2

14) Two organ pipes, each closed at one end, give 6 beats per sec, when emitting their fundamental nodes. If their lengths are in the ratio of 6 : 5, their fundamental frequencies (in Hz) are f_1 and f_2 then find the value of $f_1 + f_2$:-

(A) 66

(B) 62

(C) 68

(D) 64

15) **Statement–1:** In a Young's double slit experiment, interference pattern disappears when one of the slits is closed.
and
Statement–2: Interference occurs due to superimposition of light wave from two coherent sources.

(A) Statement–1 is True, Statement–2 is True ; Statement–2 is a correct explanation for Statement–1.

(B) Statement–1 is True, Statement–2 is True ; Statement–2 is not a correct explanation for Statement–1.

(C) Statement–1 is True, Statement–2 is False.

(D) Statement–1 is False, Statement–2 is True

16) The ratio of speed of sound in nitrogen gas to that in helium gas at 300 K is :

(A) $\sqrt{2/7}$

(B) $\sqrt{1/7}$

(C) $\sqrt{3}/5$

(D) $\sqrt{6/5}$

17) In Young's double slit experiment, one of the slit is wider than other, so that the amplitude of the light from one slit is double of that from other slit. If I_m be the maximum intensity, the resultant intensity I when they interfere at phase difference ϕ is given by :

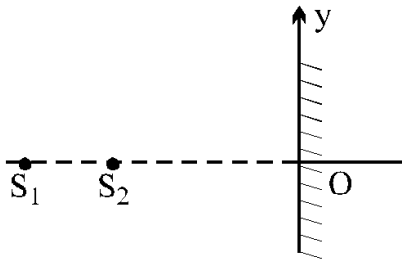
(A) $\frac{I_m}{9}(1 + 8\cos^2\frac{\phi}{2})$

(B) $\frac{I_m}{9}(4 + 5\cos\phi)$

(C) $\frac{I_m}{3}(1 + 2\cos^2\frac{\phi}{2})$

(D) $\frac{I_m}{5}(1 + 4\cos^2\frac{\phi}{2})$

18) Two point monochromatic and coherent sources of light of wavelength λ are placed on the dotted line in front of a large screen. The source emit waves in phase with each other. The distance between S_1 and S_2 is 'd' while their distance from the screen is much larger. Then,



(1) If d = 7λ/2, O will be a minima

(2) If d = 4.3λ, there will be a total of 8 minima on y axis.

(3) If d = 7λ, O will be a maxima.

(4) If d = λ, there will be only one maxima on the screen.

Which is the set of correct statement?

(A) 1, 2 & 3

(B) 2, 3 & 4

(C) 1, 2, 3 & 4

(D) 1, 3 & 4

19) The magnetic field of a plane electromagnetic wave is

$\vec{B} = 3 \times 10^{-8} \sin[200\pi(y + ct)]\hat{i}$ T

Where c = 3×10^8 ms^{-1} is the speed of light. The corresponding electric field is :

(A) $\vec{E} = -10^{-6} \sin[200\pi(y + ct)]\hat{k}$ V/m

(B) $\vec{E} = -9 \sin[200\pi(y + ct)]\hat{k}$ V/m

(C) $\vec{E} = 9 \sin[200\pi(y + ct)]\hat{k}$ V/m

(D) $\vec{E} = 3 \times 10^{-8} \sin[200\pi(y + ct)]\hat{k}$ V/m

20) In an electromagnetic wave-

(A) Power is transmitted along the magnetic field

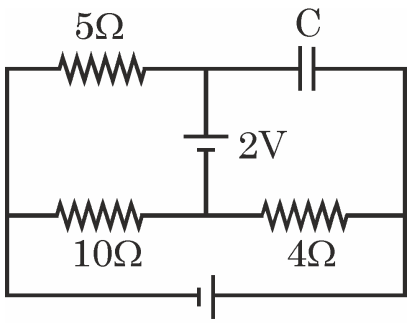
(B) Power is transmitted along the electric field

(C) Power is equally transferred along the electric and magnetic fields

(D) Power is transmitted in a direction perpendicular to both the fields

SECTION-II

1) Calculate the potential difference across the capacitor in the circuit (in Volts). (Answer nearest



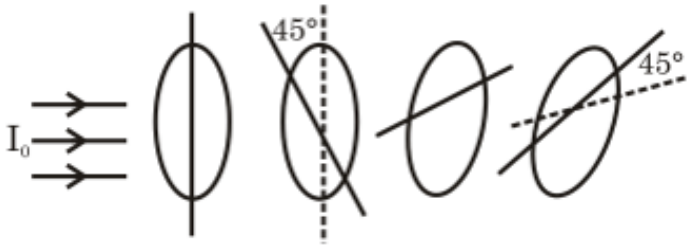
integer)

2) A wire having a linear mass density 9.0×10^{-4} kg/m is stretched between two rigid supports with a tension of 900 N. The wire resonates at a frequency of 500 Hz. The next higher frequency at which the same wire resonates is 550 Hz. The length of the wire is _____m.

3) An organ pipe 40 cm long is open at both ends. The speed of sound in air is 360 ms^{-1}. The frequency of the second harmonic is ____Hz.

4) Light of wavelength 6000Å is incident normally on a slit of width 24×10^{-5} cm. Find out the angular position (in degree) of second minimum from central maximum ?

5) Unpolarized light of intensity I_0 = 32 W/m^2 is passed through four polarizers whose transmission axis are 45º w.r.t. next polarizer. What will be output intensity in W/m^2 from last polarization ?



PART-2 : CHEMISTRY

SECTION-I

1) Choose the correct option regarding Cl-O bond length in $NaClO_4$.

(A) All Cl-O bonds are equal length.

(B) Three Cl-O bonds are of equal to length and one is longer.

(C) Two Cl-O bonds are of same length which are longer as compared to the other two Cl-O bond length.

(D) All are different.

2) In which of the following metal carbonyls the C-O bond order is lowest?

(A) $[V(CO)_6]^-$

(B) $[Mn(CO)_6]^+$

(C) $[Cr(CO)_6]$

(D) $[Ti(CO)_6]^{2-}$

3) Isomerism exhibited by $[Cr(NH_3)_2(H_2O)_2Cl_2]^+$ are.

(A) Ionisation , optical

(B) Hydrate, optical

(C) Geometrical, optical

(D) Coordinate, geometrical

4) CO_3^{2-} and HCO_3^- can be distinguished by

(A) Phenolphthalein

(B) $BaCl_2$ solution

(C) $HgCl_2$ solution

(D) All of these

5) When NH_4OH is added in $Hg_2(NO_3)_2$ solution, the ppt. formed is

(A) Hg_2O

(B) Hg + HgO

(C) Hg + $HgO.Hg(NH_2)NO_3$

(D) $HgO.Hg(NH_2)NO_3$

6) An aqueous blue coloured solution of a transition metal sulphate reacts with H_2S in acidic medium to give a black precipitate (A) which is insoluble in warm aqueous solution of KOH. The blue solution on treatment with KI in weakly acidic medium turns yellow and produces a white precipitate (B). Identify the transition metal ion.

(A) Cu

(B) Pb^{2+}

(C) Mn^{2+}

(D) Cu^{2+}

7) 0.5 F of electricity is passed through 500 mL of copper sulphate solution. The amount of copper which can be deposited will be : (Cu = 63.5)

(A) 63.5 g

(B) 31.75 g

(C) 15.875 g

(D) 254 g

8) $E^{\ominus}$ of Mg^{2+} | Mg, Zn^{2+} | Zn, and Fe^{2+} | Fe are –2.37 V, –0.76 V, and –0.44 V, respectively. Which of the following is correct ?

(A) Mg^{2+} oxidizes Fe

(B) Zn^{2+} oxidizes Fe

(C) Zn reduces Mg^{2+}

(D) Zn reduces Fe^{2+}

9) Rust (hydrated ferric oxide) may be considered as a mixture of :

(A) FeO and $Fe(OH)_2$

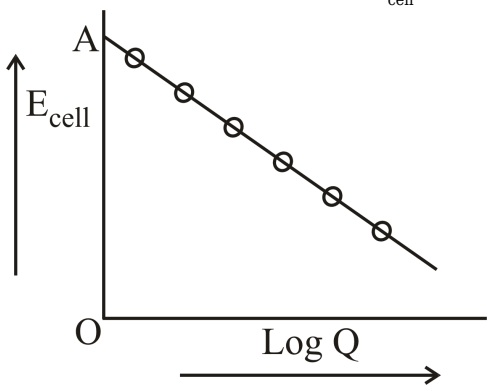
(B) FeO and $Fe(OH)_3$

(C) Fe_2O_3 and $Fe(OH)_3$

(D) Fe_3O_4 and $Fe(OH)_3$

10) $Zn(s) + Cu^{2+}(aq) \rightleftharpoons Cu(s) + Zn^{2+}(aq)$.

Reaction quotient is $Q = \frac{[Zn^{2+}]}{[Cu^{2+}]}$. Variation of E_{cell} with log Q is of the type with OA = 1.10 V. E_{cell} will



be 1.159 V when :

(A) $\frac{[Cu^{2+}]}{[Zn^{2+}]} = 0.01$

(B) $\frac{[Zn^{2+}]}{[Cu^{2+}]} = 0.01$

(C) $\frac{[Zn^{2+}]}{[Cu^{2+}]} = 0.1$

(D) $\frac{[Zn^{2+}]}{[Cu^{2+}]} = 100$

11) **Statement-1 :** HNO_3 acts only as an oxidising agent, while HNO_2 acts both as an oxidising agent and a reducing agent.
Statement-2 : The oxidation number of N in HNO_3 is maximum but oxidation number of N is in intermediate state in HNO_2.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

12) In an experiment, 6.675 g of $AlCl_3$ was produced and 0.54 g Al remained unreacted. How many g atoms of Al and Cl_2, respectively, were taken originally (Al = 27, Cl = 35.5) ?

(A) 0.075, 0.15

(B) 0.075, 0.075

(C) 0.02, 0.075

(D) 0.05, 0.075

13) Manganese ions (Mn^{2+}) can be oxidized by persulphate ions ($S_2O_8^{2-}$) according to the following equations :

$Mn^{2+} + S_2O_8^{2-} \rightarrow SO_4^{2-} + MnO_4^-$ (Acidic medium)

How many moles of $S_2O_8^{2-}$ are required to oxidise 1 mole of Mn^{2+} ?

(A) 2.5

(B) 2.0

(C) 1.0

(D) 0.4

14) Match the List :-

List-I		List-II	
(P)	Dimer of $BeCl_2$	(1)	Neither lone pair nor vacant p-orbital in valance shell of central atom
(Q)	Dimer of ABr_3	(2)	Non-planar and see-saw in shape
(R)	XeF_5^-	(3)	Central atom(s) is/are electron deficient
(S)	SF_4	(4)	Bond angles are 72°

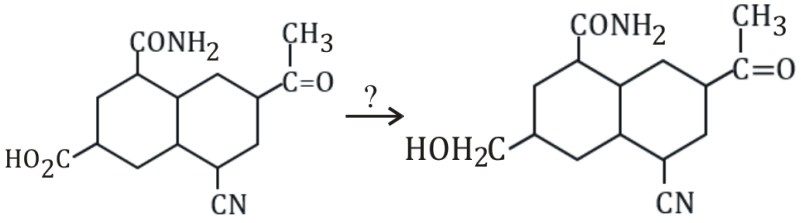
(A) P → 1 ; Q → 2 , R → 3 ; S → 4

(B) P → 2 ; Q → 3 , R → 4 ; S → 1

(C) P → 2 ; Q → 1 , R → 4 ; S → 3

(D) P → 3 ; Q → 1 , R → 4 ; S → 2

15) The most suitable reagent for the given conversion is :



(A) $LiAlH_4$

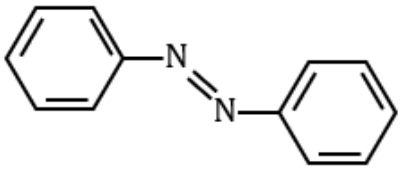
(B) $NaBH_4$

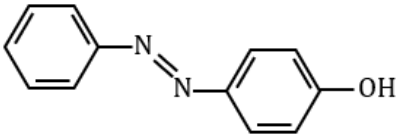
(C) H_2/Pd

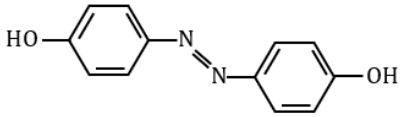
(D) B_2H_6

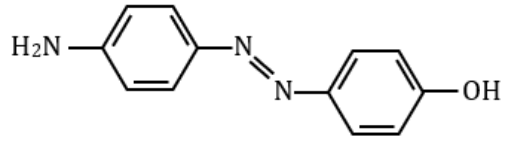
16) The azo dye obtained on reacting 4-aminophenol with nitrous acid (in dilute hydrochloric acid)

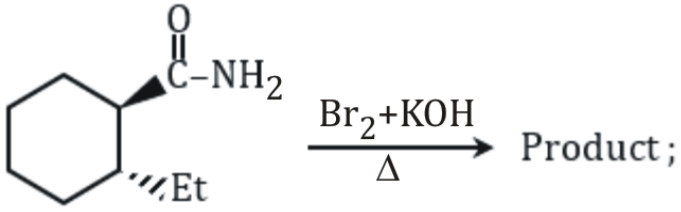
below 10°C and coupling the resulting diazonium salt with phenol is:

(A) 

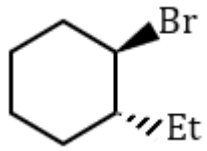
(B) 

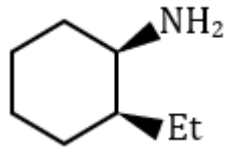
(C) 

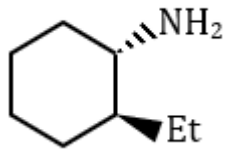
(D) 

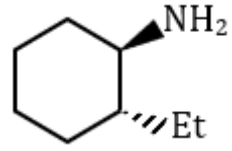
17) 

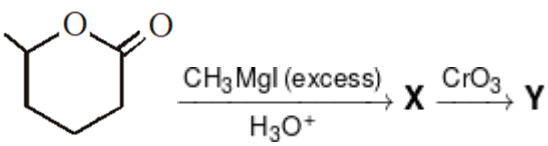
Major product is :

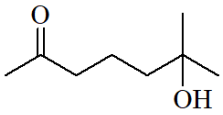
(A) 

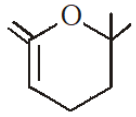
(B) 

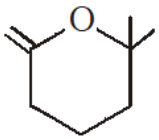
(C) 

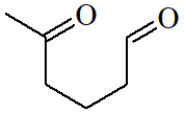
(D) 

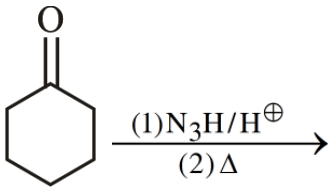
18) The major product (Y) of the following reaction is - 

(A) 

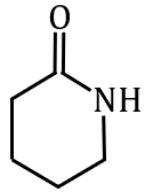
(B) 

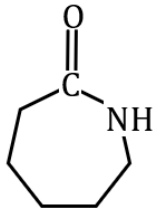
(C) 

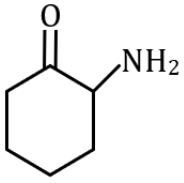
(D) 

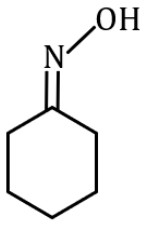
19) 

product is

(A) 

(B) 

(C) 

(D) 

20) Given below are two statements:
Statement (I): The boiling points of alcohols and phenols increase with increase in the number of C-atoms.
Statement (II): The boiling points of alcohols and phenols are higher in comparison to other class of compounds such as ethers, haloalkanes.
In the light of the above statements, choose the **correct** answer from the options given below:

(A) Both **Statement I** and **Statement II** are false

(B) **Statement I** is false but **Statement II** is true

(C) **Statement I** is true but **Statement II** is false

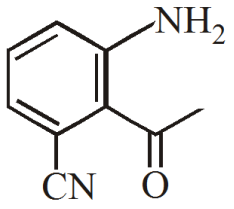
(D) Both **Statement I** and **Statement II** are true

SECTION-II

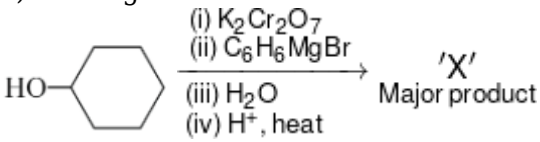
1) How many of the following ions can form coloured complexes?
Cu^{2+} , Zn^{2+} , Ti^{3+} , Ti^{4+} , Cd^{2+} and Mn^{2+}

2) If the following redox reaction is balanced with smallest whole number coefficient
$Cr_2O_7^{2-} + H^+ + Ni \rightarrow Cr^{3+} + Ni^{2+}$,
the stoichiometric coefficient of Ni in the balanced reaction is.

3) Equivalent conductance of 0.2 M aqueous solution of a weak monobasic acid (HA) is 10 S cm^2 $equiv^{-1}$ and that at infinite dilution is 200 S cm^2 $equiv^{-1}$. Hence, pH of this solution is.

4) The lone pair of electrons in (A) is : 

(A)

5) In the given reaction


the number of sp^2 hybridised carbon (s) in compound 'X' is____.

PART-3 : MATHEMATICS

SECTION-I

1) If rolle's theorem is applicable on $f(x) = \begin{cases} ax+b & ; -\infty < x \leq 2 \\ x^2-5x+6 & ; 2 < x < 3 \\ px^2+qx+1 & ; 3 \leq x < \infty \end{cases}$, then

(A) $a = -1 ; p = -\frac{5}{3}$

(B) $b = 2 ; p = \frac{4}{9} ; q = \frac{5}{3}$

(C) $b = 2 ; p = \frac{4}{9} ; q = -\frac{5}{3}$

(D) $a = 1 ; p = \frac{4}{9} ; q = -\frac{5}{3}$

2) Let $f(x) = \begin{cases} -x^2 + \log_2 a & , \ 0 \leq x < 1 \\ 3x & , \ 1 \leq x \leq 3 \end{cases}$. The complete set of real values of 'a' for which f(x) has smallest value at x = 1, is

(A) (0, 8]

(B) [8, ∞)

(C) (0, 16]

(D) [16, ∞)

3) Consider the differential equation $(xy - 1)\frac{dy}{dx} + y^2 = 0$

Assertion A : The solution of the equation is xy = logy + c.

Reason R : Equation can be expressed as linear differential equation with integrating factor y.

(A) **A** is correct but **R** is not correct

(B) **A** is not correct but **R** is correct

(C) Both **A** and **R** are correct but **R** is **NOT** the correct explanation of **A**

(D) Both **A** and **R** are correct and **R** is the correct explanation of **A**

4) Number of integral values of p for which the cubic equation $2x^3 - 3x^2 + p = 0$ has 3 real roots, is

(A) 1

(B) 2

(C) 3

(D) 4

5) Equation of normal from (5,5) to the curve represented parametrically y = 2t and $x = \frac{2}{t}$ is

(A) y = x + 2

(B) y = 2x – 1

(C) y = x

(D) y = 3x – 5

6) $\int \frac{x^5 dx}{(x^5 + x)^2}$ is equal to

(Where c is constant of integration)

(A) $\frac{x^4+1}{4x^4}+c$

(B) $\frac{x^4+1}{2(x^2+1)}+c$

(C) $\frac{-1}{4(x^4+1)}+c$

(D) $\frac{x^2+1}{2(x^4+1)}+c$

7) $\int e^{\tan^{-1}x}(1+x+x^2).d(\cot^{-1}x)$ is equal to

(A) $-e^{\tan^{-1}x}+c$

(B) $e^{\tan^{-1}x}+c$

(C) $-x.e^{\tan^{-1}x}+c$

(D) $x.e^{\tan^{-1}x}+c$

8) $\int e^x\left(\frac{x^2+7x+13}{(x+5)^2}\right)dx$ is equal to

(A) $e^x\left(\frac{x+2}{x+5}\right)+C$

(B) $e^x\left(\frac{x+3}{x+5}\right)+C$

(C) $e^x\left(\frac{x-2}{x+5}\right)+C$

(D) $e^x\left(\frac{x-3}{x+5}\right)+C$

(where C is constant of integration)

9) Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(A)	Value of $\int_0^1 \frac{dx}{\sqrt{1-x^2}\left(1+\sqrt{1-x^2}\right)}$ is equal to	(I)	2
(B)	$f: R \to R$ defined as $f(x)=x^5+e^{\frac{x}{2}}+e^{x^3}$ & $g(x)=f^{-1}(x)$, then value of $g'(2)$ is equal to	(II)	4
(C)	$\lim_{n\to\infty}\frac{1}{\sqrt{n}}\left(1+\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{3}}+\ldots\frac{1}{2\sqrt{n}}\right)$ equal to	(III)	12

(D)	If $f(x)=\int_{x-1}^{2x}\left(t^3-2t+4\right)dt$, then value of f ′ (1) is equal to	(IV)	1

(A) (A) – IV, (B) – I, (C) – II, (D) – III

(B) (A) – I, (B) – II, (C) – IV, (D) – III

(C) (A) – II, (B) – I, (C) – III, (D) – IV

(D) (A) – III, (B) – IV, (C) – II, (D) – I

10) The value of $\int_{-1/\sqrt{3}}^{1/\sqrt{3}}\frac{x^4}{1-x^4}\cos^{-1}\left(\frac{2x}{1+x^2}\right)dx$ is

(A) $\frac{\pi}{12}\left(\pi+3\log\left(2+\sqrt{3}\right)-4\sqrt{3}\right)$

(B) $\frac{\pi}{12}\left(\pi-3\log\left(2+\sqrt{3}\right)-4\sqrt{3}\right)$

(C) $\frac{\pi}{12}\left(\pi+3\log\left(2+\sqrt{3}\right)+4\sqrt{3}\right)$

(D) $\frac{\pi}{12}\left(\pi-3\log\left(2+\sqrt{3}\right)+4\sqrt{3}\right)$

11) If $\int_0^x f(t)\,dt=x^2+\int_x^1 t^2 f(t)\,dt$, then $f'\left(\frac{1}{2}\right)$ is :

(A) $\frac{6}{25}$

(B) $\frac{24}{25}$

(C) $\frac{18}{25}$

(D) $\frac{4}{5}$

12) $\int_0^{\pi/2}\frac{\left(\sin^{10}x+\tan^7x\right)}{\left(\sin^{10}x+\cos^{10}x+\tan^7x+\cot^7x\right)}dx$ is

(A) π

(B) $\frac{\pi}{2}$

(C) $\frac{\pi}{8}$

(D) $\frac{\pi}{4}$

13) $\int_{a+c}^{b+c} f(x)\,dx$ is equal to -

(A) $\int_a^b f(x+c)\,dx$

(B) $\int_a^b f(x)\,dx$

(C) $\int_{a-c}^{b-c} f(x-c)\,dx$

(D) $\int_a^b f(x-c)\,dx$

14) Area of region enclosed between curves given by $y^2 = -4x$ and $-x = \sqrt{|y|}$ is (in sq. units)-

(A) $\frac{1}{3}$

(B) $\frac{2}{3}$

(C) $\frac{4}{3}$

(D) $\frac{8}{3}$

15) Let f be a function such that $f'(x) > 0\ \forall\ x \in R^+$

Statement-I : The equation $4f(x) = \int_0^4 f(x)\,dx$ has exactly one solution in the interval (0, 4)

Statement-II : If f(x) is a continuous function in [a, b] then $(b - a)m \le \int_a^b f(x)\,dx \le (b - a)M$ where m and M be the minimum and maximum values of 'f' respectively in [a, b]

(A) Statement-I is true but Statement-II is false

(B) Both Statement-I and Statement-II are false

(C) Both Statement-I and Statement-II are true

(D) Statement-I is false but Statement-II is true

16) Area enclosed by $x^2 = 4y$ and $y = \frac{8}{x^2 + 4}$ is

(A) $\pi - \frac{2}{3}$

(B) $\pi - \frac{1}{3}$

(C) $2\pi - \frac{4}{3}$

(D) $2\pi - \frac{2}{3}$

17) The area (in sq. units) of the region bounded by the curves $y = 2^x$ and $y = |x + 1|$, in the first quadrant is :

(A) $\frac{3}{2} - \frac{1}{\log_e 2}$

(B) $\frac{1}{2}$

(C) $\log_e 2 + \frac{3}{2}$

(D) $\frac{3}{2}$

18) The solution of differentiable equation $2y \sin x\frac{dy}{dx} = \sin 2x - y^2 \cos x$ satisfying $y\left(\frac{\pi}{2}\right) = 1$ is -

(A) $y^2 = \cos x + 1$

(B) $y = \sin^2 x$

(C) $y^2 \sin x = 4 \cos^2 x + 1$

(D) $y^2 = \sin x$

19) The solution of differential equation $\frac{dy}{dx} = \frac{6x^2y}{2x^3 + 5y^4}$ is:

(A) $2x^3y^{-1} = \frac{5y^3}{3} + C$

(B) $2x^2y^{-1} = \frac{5}{3}y^3 + C$

(C) $2xy^{-1} = \frac{5}{3}y^3 + C$

(D) $2x^4y^{-1} = \frac{5}{3}y^3 + C$

20) General solution of the differential equation $f(x)\frac{dy}{dx} = f^2(x) + f(x)y + f'(x)y$ is
(Where c is the constant of integration)

(A) $y = f(x) + ce^{-x}$

(B) $y = -f(x) + ce^x f(x)$

(C) $y = f(x) - ce^x$

(D) $y = f(x) - ce^{-x}$

SECTION-II

1) If y(x) satisfy the differential equation $\left(x^2 - 1\right)\frac{dy}{dx} + 2xy = 1$ where y(0) = –10 then y(2) is equal to

2) If $f(x) = \begin{vmatrix} \cos x & e^{x^2} & 3x \\ x^4 & \cos x & \tan x + x \\ 3 & 1 & x^3 \end{vmatrix}$, then the value of $\int_{-\pi/2}^{\pi/2} \left(x^2 + 1\right)\left(f(x) + f''(x)\right) dx$ is k then [k] is equal to

(Note : [x] denotes greatest integer less than or equal to x)

3) Let $\int \frac{(5x^4 + 1)dx}{3x^{4/3}\sqrt{x^4 - 1}} = \frac{\sqrt{f(x)}}{x^m} + C$ (where $f(0) = -1$), then $\frac{3}{m}$ is equal to

4) Let $f(x) = \begin{cases} 1 & x = 0 \\ x^2 - a & 0 < x < 1 \\ bx & 1 \leqslant x < 2 \\ 6 - c & x = 2 \end{cases}$ be such that Lagrange's mean value theorem is applicable to f(x) for $x \in [0, 2]$. Let P(p, f(p)), $p \in (0, 2)$ be a point on f(x) such that tangent at P is parallel to chord joining points with x = 0 & x = 2 on y = f(x). Then a + b + c + 4p is equal to

5) Area bounded by the straight lines $x^2y - y^3 - x^2 + 5y^2 - 8y + 4 = 0$ is equal to (in sq. units)

ANSWER KEYS

PART-1 : PHYSICS

SECTION-I

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	B	C	D	D	B	D	B	D	B	C	C	B	D	A	A	C	A	C	B	D

SECTION-II

Q.	21	22	23	24	25
A.	3	10	900	30	2

PART-2 : CHEMISTRY

SECTION-I

Q.	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
A.	A	D	C	D	C	D	C	D	C	B	A	B	A	D	D	C	D	A	B	D

SECTION-II

Q.	46	47	48	49	50
A.	3	3	2	4	8

PART-3 : MATHEMATICS

SECTION-I

Q.	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70
A.	C	D	C	B	C	C	C	A	A	A	B	D	A	D	C	C	A	D	A	B

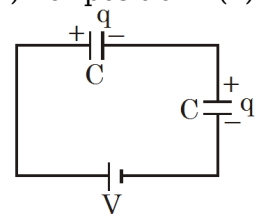
SECTION-II

Q.	71	72	73	74	75
A.	4	0	9	6	1

SOLUTIONS

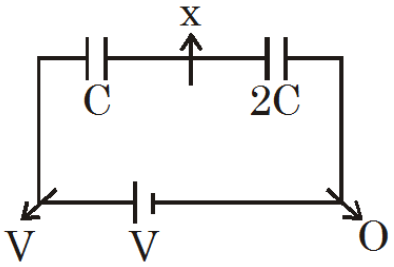
PART-1 : PHYSICS

1) For position---(1)



$q = \frac{C}{2}v$

For position---(2)



$(x - V)C + (x - 0)2C = -q = -\frac{CV}{2}$

$3x - V = -\frac{V}{2}$

$x = \frac{V}{6}$

$q_{2C} = 2C(x - 0) = \frac{CV}{3}$

3) Series - parallel combination.

4)

Here $R_{eq} = R' = \frac{7R}{4}$

Now $i = \frac{q_0}{R'C}e^{-\frac{t}{R'C}}$

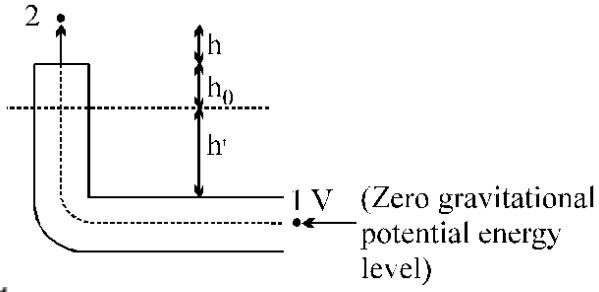
$\frac{1}{4}\frac{q_0}{R'C} = \frac{q_0}{R'C}e^{-\frac{t}{R'C}}$

or $\ln 4 = \frac{t}{R'C}$

$t = \frac{7RC}{4}\ell n4 = \frac{7RC}{2}\ell n2$

5) Applying Bernoulis theorem for points 1 and 2 of the streamline shown

P_0 → Atmospheric pressure



$\frac{1}{2}\rho V^2 + (P_0 + \rho g h') + 0$

$= 0 + P_0 + \rho g(h' + h_0 + h)$

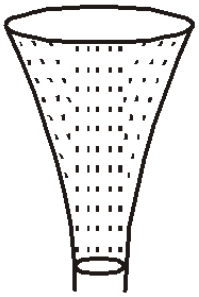
$\Rightarrow h = 10$ cm

7)

$VT \propto r^2$

9) Conceptual

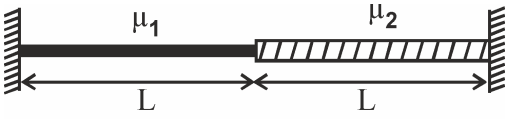
10) Pressure is atmospheric



11) Compare with

$$y = A\cos\left[2\pi\left(\frac{t}{T} - \frac{x}{\lambda}\right)\right]$$

$\lambda = vt$ & $v = \sqrt{\frac{F}{\mu}}$



13) radius = r radius = 2r

Let mass per unit length of wires are μ_1 and μ_2 respectively.

$\because$ Materials are same, so density ρ is same.

$\therefore \mu_1 = \frac{\rho\pi r^2 L}{L} = \mu$ and $\mu_2 = \frac{\rho 4\pi r^2 L}{L} = 4\mu$

Tension in both are same = T, let speed of wave in wires are V_1 and V_2

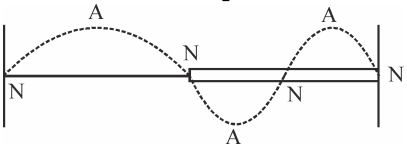
$V_1 = \sqrt{\frac{T}{\mu}} = V$; $V_2 = \sqrt{\frac{T}{4\mu}} = \frac{V}{2}$

So fundamental frequencies in both wires are

$$f_{01} = \frac{V_1}{2L} = \frac{V}{2L} \text{ \& } f_{02} = \frac{V_2}{2L} = \frac{V}{4L}$$

Frequency at which both resonate is L.C.M of both frequencies i.e. $\frac{V}{2L}$.

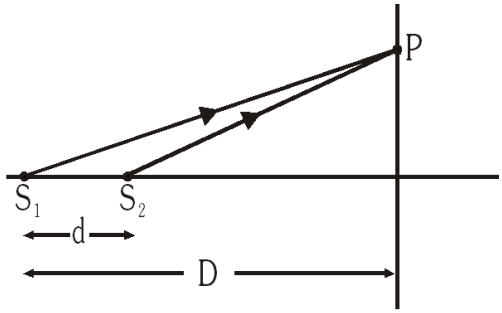
Hence no. of loops in wires are 1 and 2 respectively.



So, ratio of no. of antinodes is 1 : 2.

17) For bright fringe $S_1P - S_2P = n\lambda$

So fringes are concentric circles (centre of origin)



19) $\vec{B} = 3 \times 10^{-8} \sin[200\pi(y + ct)]\hat{i}\ T$

$E_0 = CB_0 \Rightarrow E_0 = 3 \times 10^8 \times 3 \times 10^{-8} = 9$ V/m

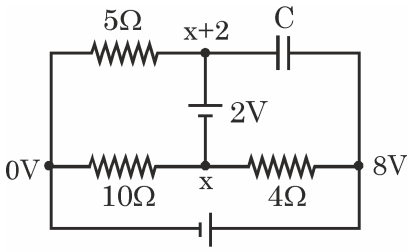
and direction of wave propagation is given as

$\left(\vec{E} \times \vec{B}\right) \| \vec{C}$

$\hat{B} = \hat{i}$ & $\hat{C} = -\hat{j}$

so $\hat{E} = -\hat{k}$

$\therefore \vec{E} = E_0 \sin[200\pi(y + ct)](-\hat{k})$ V/m



21)

Net current at point x = 0.

$$\therefore \frac{0-x}{10} + \frac{8-x}{4} + \frac{0-(x+2)}{5} = 0$$

Potential across capacitor ⇒ [(x + 2) – 8]

22) $\mu = 9.0 \times 10^{-4} \frac{kg}{m}$

T = 900 N

$$V = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{900}{9 \times 10^{-4}}} = 1000 m/s$$

$f_1 = 500$ Hz

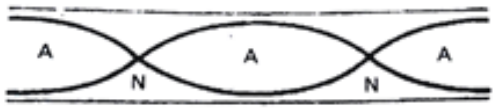
$f = 550$

$$\frac{nV}{2\ell} = 500 \quad \text{.... (i)}$$

$$\frac{(n+1)V}{2\ell} = 500 \quad \text{.... (ii)}$$

(ii) (i) $\frac{V}{2\ell} = 50$

$$\ell = \frac{1000}{2 \times 50} = 10$$

23) 

For second harmonic of open organ pipe

$L = \lambda$

So frequency of vibration is $f = \frac{V}{\lambda}$

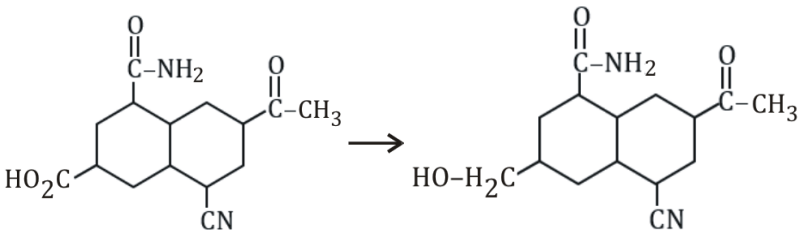
$$f = \frac{V}{\lambda} = \frac{V}{L} = \frac{360}{\frac{40}{100}} = 900\,Hz$$

25) $$I_{out} = \left(\frac{I_0}{2}\right)\cos^2(45) \times \cos^2(45)\cos^2(45)$$

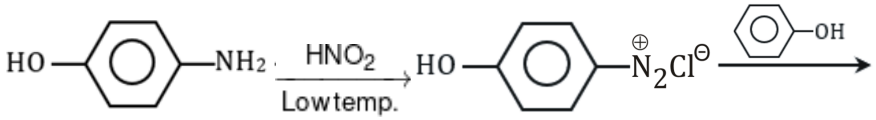
$$I_{out} = \frac{I_0}{2^4} = \frac{I_0}{16} = 2\,w/m^2$$

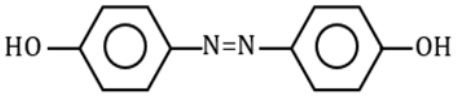
PART-2 : CHEMISTRY

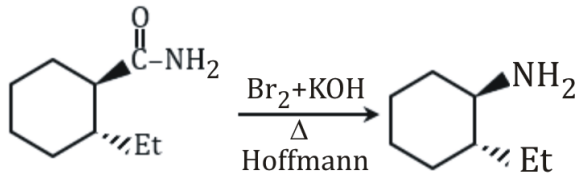
39) On the basis of bridge bonding & dimerisation

40) 

Most suitable reagent for given conversion is B_2H_6 (electrophilic reducing agent)

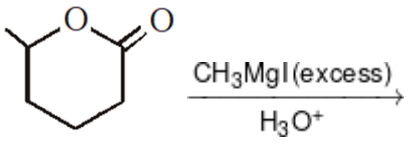
41) 

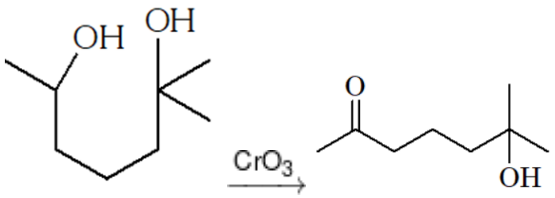


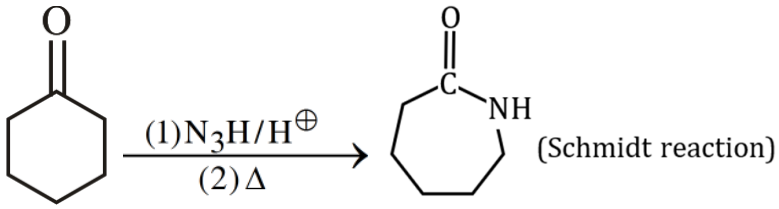


42)

In Hoffmann Bromamide retention takes place.

43) 



44) 

(Schmidt reaction)

45) B.P. ∝ M.W.

B.P. ∝ Inter molecular hydrogen bonding

Alcohol & Phenol have intermolecular H-bonding

46) $Cu^{2+} \rightarrow 3d^9$

$Zn^{2+} \rightarrow 3d^{10}$

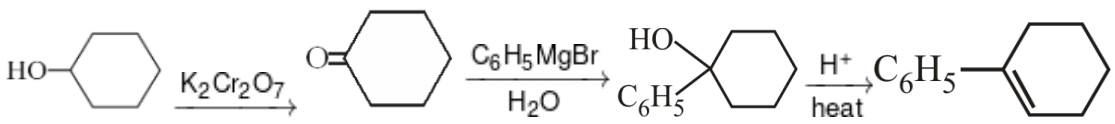
$Ti^{3+} \rightarrow 3d^1$

$Ti^{4+} \rightarrow 3d^0$

$Cd^{2+} \rightarrow 4d^{10}$

$Mn^{2+} \rightarrow 3d^5$

$3d^9$, $3d^1$ and $3d^5$ can show coloured complexes.

50) 

PART-3 : MATHEMATICS

51) $a = -1$; $p = \frac{4}{9}$, $q = -\frac{5}{3}$, $b = 2$

52) $\lim_{x \to 1} f(x) \geq f(1) \Rightarrow a \geq 16$

53) Given equation $xydy + y^2dx - dy = 0$

$$y^2\frac{dx}{dy} + xy = 1$$

$$\frac{dx}{dy} + \frac{x}{y} = \frac{1}{y^2}$$

This is a linear differential equation whose integrating factor is $e^{\int \frac{1}{y}dy} = y$ and solution will be

$$xy = \int \frac{1}{y}dy$$

$xy = \log y + c$

hence **Assertion A** is true and **Reason R** is false.

54) Ans. (2)

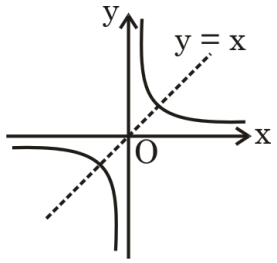
Let $f(x)\ 2x^3 - 3x^2 + p$

$f'(x) = bx(x - 1)$

$f(x).f(1) < 0$ (Not necesserily distrinct)

$p \in [0, 1]$

55)



56) $$\int \frac{x^5dx}{(x^5 + x)^2} = \int \frac{dx}{x^5\left(1 + \frac{1}{x^4}\right)^2}$$

Now take $1 + \frac{1}{x^4} = t$

57) $$\int e^{\tan^{-1}x}(1 + x + x^2).d(\cot^{-1}x)$$

$$\int e^{\tan^{-1}x}(1 + x + x^2)\frac{-1}{1 + x^2}dx$$

$$-\int e^{\tan^{-1}x}(1 + x^2 + x)\frac{-1}{(1 + x^2)}dx$$

$$-\int e^{\tan^{-1}x}(1+\frac{x}{1+x^2}dx$$
$$-\int e^{\tan^{-1}x}.dx-\int e^{\tan^{-1}x}\frac{x}{1+x^2}dx$$
$$-e^{\tan^{-1}x}.x+\int e^{\tan^{-1}x}\frac{1}{1+x^2}.x\,dx-\int\frac{e^{\tan^{-1}x}}{1+x^2}x\,dx$$
$$=-x\,e^{\tan^{-1}x}+C$$
Option (C) is correct.

58) $\int e^x\left(\frac{x^2+7x+10+3}{(x+5)^2}\right)dx$

$$\Rightarrow\int e^x\left(\frac{(x+2)(x+5)+3}{(x+5)^2}\right)dx$$
$$\Rightarrow\int e^x\left(\frac{x+2}{x+5}+\frac{3}{(x+5)^2}\right)dx$$
$$\Rightarrow\int e^x\left(f(x)+f'(x)\right)dx$$
$$=e^x\left(\frac{x+2}{x+5}\right)+C$$

59) (A) Put $x=\sin\theta$

$$\int_0^{\pi/2}\frac{d\theta}{1+\cos\theta}=1$$

(B) $f'(x)=5x^4+\frac{1}{2}e^{x/2}+3x^2e^{x^3}>0$

$\therefore$ f is one-one map

$gf(x)=x \Rightarrow g'(f(x)).f'(x)=1$

$$g(f(x))=\frac{1}{f'(x)}$$

& $f(0)=2$

$\therefore g'(2)=2$

(C) $T_r=\frac{1}{\sqrt{n}\sqrt{r}}=\frac{1}{n(\sqrt{r/n})}$

$$\text{Sum}=\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{4n}\frac{1}{\sqrt{r/n}}=\int_0^4\frac{1}{\sqrt{x}}dx=4$$

(D) $f'(x)=[(2x)^3-2(2x)+4]\times 2$
$-[(x-1)^3-2(x-1)+4]$

$f'(1)=12$

60) $\int_{-1/\sqrt{3}}^{1/\sqrt{3}}\frac{x^4}{1-x^4}\cos^{-1}\left(\frac{2x}{1+x^2}\right)dx$

Let $\tan^{-1}x=\theta$

$x=\tan\theta$

$dx = \sec^2\theta\, d\theta$

$$\int_{-\pi/6}^{\pi/6} \frac{\tan^4\theta}{1-\tan^4\theta}\cos^{-1}(\sin 2\theta)\, d\theta \sec^2\theta$$

$$\int_{-\pi/6}^{\pi/6} \frac{\tan^4\theta}{1-\tan^4\theta}\left(\frac{\pi}{2}-\sin^{-1}(\sin 2\theta)\right) d\theta \sec^2\theta$$

$$\int_{-\pi/6}^{\pi/6} \frac{\tan^4\theta}{\left(1-\tan^4\theta\right)}\left(\frac{\pi}{2}-2\theta\right) d\theta$$

$$2\int_{0}^{\pi/6} \frac{\pi}{2}\frac{\tan^4\theta}{1-\tan^2\theta} d\theta$$

$$= \frac{\pi}{12}\left[\pi + 3\log\left(2+\sqrt{3}\right) - 4\sqrt{3}\right]$$

61) $\int_0^x f(t)\,dt = x^2 + \int_x^1 t^2 f(t)\,dt \qquad f'\left(\frac{1}{2}\right) = ?$

Differentiate w.r.t. 'x'

$f(x) = 2x + 0 - x^2 f(x)$

$$f(x) = \frac{2x}{1+x^2}$$

$$f'(x) = \frac{(1+x^2)2 - 2x(2x)}{(1+x^2)^2}$$

$$f'(x) = \frac{2x^2 - 4x^2 + 2}{(1+x^2)^2}$$

$$f'\left(\frac{1}{2}\right) = \frac{2-2\left(\frac{1}{4}\right)}{\left(1+\frac{1}{4}\right)^2} = \frac{\left(\frac{3}{2}\right)}{\frac{25}{16}} = \frac{48}{50} = \frac{24}{25}$$

.

62) $$I = \int_0^{\pi/2} \frac{\left(\sin^{10}x + \tan^7 x\right)}{\left(\sin^{10}x + \cos^{10}x + \tan^7 x + \cot^7 x\right)} dx \quad \ldots\ldots(1)$$

$$I = \int_0^{\pi/2} \frac{\left(\cos^{10}x + \cot^7 x\right)}{\left(\sin^{10}x + \cos^{10}x + \tan^7 x + \cot^7 x\right)} dx$$

$$2I = \int_0^{\pi/2} dx = \frac{\pi}{2}$$

(1) + (2) ⇒

$$I = \frac{\pi}{4}$$

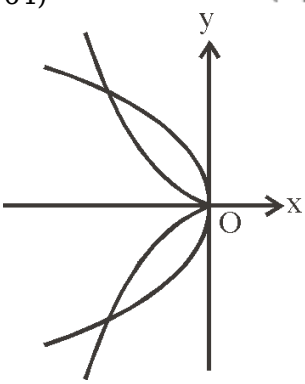
63) $\int_{a+c}^{b+c} f(x)dx$

put x = t + c

$$\int_a^b f(t+c)dt$$

$$\int_a^b f(x+c)dx$$

64) $A = 2 \times \frac{16}{3}(1)\left(\frac{1}{4}\right) = \frac{8}{3}$



65) **Statement-II :** is obviously true.

Statement-I : $4f(x) = \int_0^4 f(x)dx$

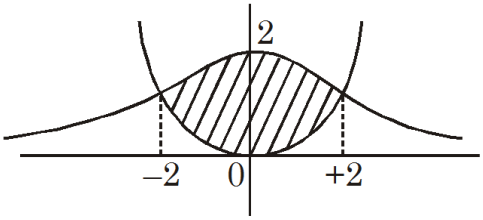
f (x) = $\frac{1}{4-0}\int_0^4 f(x)\,dx$

RHS lies in the range of f(x) (from statement-II)

 f(x) is one-one

 The given equation has exactly one solution.

66) $x^2 = 4y$ $y = \frac{8}{x^2+4}$



$\frac{8}{x^2+4} = \frac{x^2}{4} \Rightarrow x^4 + 4x^2 = 32$

$\Rightarrow x^4 + 4x^2 - 32 = 0$

$\Rightarrow (x^2 + 8)(x^2 - 4) = 0$

$\Rightarrow x = \pm 2$

 Area = $2\int_0^2 \left(\frac{8}{x^2+4} - \frac{x^2}{4}\right) dx$

$$= 2\left[8.\frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right) - \frac{x^3}{12}\right]_0^2$$
$$= 2\left[4\tan^{-1}(1) - \frac{8}{12}\right] = 2\pi - \frac{4}{3}$$

67) Area $= \int_0^1 (|1x+1| - 2^x)dx$
$= \frac{3}{2} - \frac{1}{\log_e^2}.$

68) $2y\sin x.\frac{dy}{dx} = 2\sin x\cos x - y^2\cos x$
2y sin x dy = sin 2x dy - y^2 cosx dx
2y sinx dy + y^2 cos x dx = sin 2x dx
$\int d\left(y^2\sin x\right) = \int \sin 2x\,dx$
$y^2\sin x = \frac{-\cos 2x}{2} + C$
$1.1 = \frac{1}{2} + C \Rightarrow C = \frac{1}{2}$
$y^2.\sin x = \frac{1}{2}(1 - \cos 2x)$
$y^2.\sin x = \frac{2\sin^2 x}{2}$
y^2 = sin x

69) $2x^3$ dy + $5y^4$ dy =$6x^2$ y dx
$5y^4$ dy = $6x^2$y dx - $2x^3$ dx
$5y^2dy = 2\ d\left(\frac{x^3}{y}\right)$
$\frac{5y^3}{3} = 2x^3.y^{-1} + C$

70) $\frac{dy}{dx} = \left(1 + \frac{f'(x)}{f(x)}\right)y = f(x)$
$If = e^{-\int\left(1+\frac{f'(x)}{f(x)}\right)dx} = \frac{e^{-x}}{f(x)}$
$y\left(\frac{e^{-x}}{f(x)}\right) = \int e^{-x}dx \Rightarrow \frac{ye^{-x}}{f(x)} = -e^{-x} + c$
y = -f(x) + ce^xf(x)

71) $\left(x^2-1\right)\frac{dy}{dx} + 2xy = 1 \Rightarrow \frac{d}{dx}((x^2-1)y) = 1$
(x^2 -1)y = x + c
y(0) = -10 ⇒ c = 10

$(x^2 - 1)y = 10 + x$
$y(2) = 4$

72) $f(x)$ is an odd functions.
$f'(x)$ is an even functions.
$f''(x)$ is an odd functions.
$\therefore$ $f(x) + f''(x)$ is an odd functions

Hence $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (x^2 + 1)\,(f(x) + f''(x))\,dx = 0$

73) Let $I = \int \frac{(6x^4 - (x^4 - 1)dx}{3x^{4/3}\sqrt{x^4 - 1}} =$

$\int \left(\frac{2x^{8/3}}{\sqrt{x^4 - 1}} - \frac{\sqrt{x^4 - 1}}{3x^{4/3}} \right)$

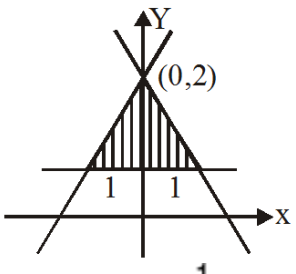
$= \int \frac{d}{dx}\left(\frac{\sqrt{x^4 - 1}}{x^{1/3}} \right) dx = \frac{\sqrt{x^4 - 1}}{x^{1/3}} + C$

$\Rightarrow \frac{1}{m} = 3 \Rightarrow \frac{3}{m} = 9$

Aliter :
Differentiate both sides to get $f(x)$.

74) Since LMVT is applicable to f (x) $\forall$ x $\in$ [0, 2]
$\therefore$ f (x) is continuous in [0, 2] and differentiable in (0, 2)
$\Rightarrow$ a = –1, b = 2, c = 2
Also $f'(p) = \frac{f(2) - f(0)}{2 - 0} = \frac{4 - 1}{2} = \frac{3}{2}$
gives $p = \frac{3}{4}$
Hence a + b + c + 4p = 6

75) $x^2y - x^2 - y^3 + y^2 + 4y^2 - 4y - 4y + 4 = 0$
$\Rightarrow x^2(y-1) - y^2(y-1) + 4y(y-1) - 4(y-1) = 0$
$\Rightarrow (y - 1)(x^2 - y^2 + 4y - 4) = 0$
$\Rightarrow$ $y = 1$ or $x^2 = (y - 2)^2$
$\Rightarrow$ $y = 1$, $x = y - 2$ or $x = 2 - y$



Shaded area = $\frac{1}{2} \times 2 \times 1 = 1$ sq. units